

Seven Segment Display Interfacing via 7447 IC with Vamon FPGA Board

Aim

To interface a seven segment display with the Vamon FPGA board through a 7447 BCD-to-seven-segment decoder IC, by sending a 4-bit BCD value from the Vamon board to the 7447 IC, and to verify on hardware that the decoder drives the correct segments to display the corresponding digit.

GATE Question

The Boolean expression for the output f of a 4:1 multiplexer with inputs $I_0 = R$, $I_1 = R'$, $I_2 = R'$, $I_3 = R$ and select lines $S_1 = P$, $S_0 = Q$ is one of the following:

Option	Expression
A	$(P \text{ xor } Q \text{ xor } R)'$
B	$P \text{ xor } Q \text{ xor } R$
C	$P + Q + R$
D	$(P + Q + R)'$

Question Analysis

1. Select-Line to Output Mapping

$S_1 (P)$	$S_0 (Q)$	Line Selected	Output f
0	0	I_0	R
0	1	I_1	R'
1	0	I_2	R'
1	1	I_3	R

2. Sum of Products Form

$$f = P'Q'R + P'Q R' + PQ' R' + PQR$$

3. Complete Truth Table

P	Q	R	f
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0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

4. Observation and Final Expression

f is 1 only where P, Q, R contain an odd number of 1's (001, 010, 100, 111) – the defining property of the three-input XOR operation.

$f = P \text{ xor } Q \text{ xor } R \rightarrow \text{Answer: Option (B)}$

1. Components Used

Component
Vamon Board
7447 BCD to Seven Segment Decoder IC
Seven Segment Display (Common Anode)
Jumper Wires
Linux Computer

2. Purpose of Each Component

Component	Function
Vamon Board	Sends a 4-bit BCD value (A, B, C, D) to the 7447 IC
7447 IC	Decodes the BCD input and drives the seven segment outputs
Seven Segment Display	Lights up the segments to show the resulting digit
Jumper Wires	Connects the Vamon board, 7447 IC, and display together
Linux Computer	Runs Termux/toolchain to compile and program the board

3. Power Connections

Pin	Connection
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7447 pin 16 (Vcc)	+5V
7447 pin 8 (GND)	GND
7447 pin 3 (LT)	+5V
7447 pin 4 (BI/RBO)	+5V
7447 pin 5 (RBI)	+5V
7-Segment Common Anode Pins	+5V

4. Vamon Board to 7447 IC Connections

Vamon Pin	7447 Pin
A	Pin 7
B	Pin 1
C	Pin 2
D	Pin 6

5. 7447 IC to Seven Segment Display Connections

7447 Pin	7-Segment Pin
13	a
12	b
11	c
10	d
9	e
15	f
14	g

6. Execution Procedure

1. In Termux on the Linux computer, write a design that outputs a 4-bit BCD value on pins A, B, C, D of the Vamon board, along with the constraint file for pin mapping.
2. Synthesize the design through the SymbiFlow toolchain and convert the resulting bitstream to a .bin file.
3. Connect the Vamon board to the computer over USB and place it into programming mode.
4. Confirm the board is recognized as a serial device, then flash the .bin file using the TinyFPGA programmer utility.
5. Reset the board so the BCD output logic takes effect.

6. Vary the BCD value on A, B, C, D and observe the seven segment display: the 7447 IC should decode each value and light the matching digit (0-9).

Results and Observations

With the 7447 IC powered and its control pins (LT, BI/RBO, RBI) held at +5V, the IC correctly decoded each 4-bit BCD value sent from the Vamon board and drove the seven segment display accordingly. The common anode pins, held at +5V, allowed the active-LOW outputs of the 7447 to turn the corresponding segments on, and the display showed the expected decimal digit for every BCD value tested.